

Cognitive Digital Twin — Nagpur Orange Supply Chain

Project report submitted in partial fulfillment of the requirements for the Final Year Project

Abstract

The supply chain of agricultural products, specifically the Nagpur orange, faces significant challenges regarding freshness, routing, and inventory tracking. This project proposes a Cognitive Digital Twin—a real-time simulated supply chain network of the Nagpur orange distribution system. By integrating artificial intelligence and a digital twin architecture, the system simulates autonomous warehouse, truck, and retailer agents. It features a PPO-based Reinforcement Learning (RL) agent for optimal truck routing, live weather and accident disruption models, and real-time inventory tracking using Extended Kalman Filters (EKF). The whole infrastructure is monitored via a live React-based dashboard connected to an InfluxDB telemetry stack. This approach provides proactive decision-making, predictive ETA, and Cargo Freshness (RSL) tracking, leading to a highly optimized and resilient supply chain.

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CHAPTER 1: INTRODUCTION

1.1 Background

The Nagpur orange, a highly perishable agricultural commodity, requires an efficient supply chain to minimize spoilage and maximize economic returns. Traditional supply chain management relies on static routing and periodic inventory reviews. However, these traditional models fail to capture the dynamic and stochastic nature of weather disruptions, traffic accidents, and rapid cargo degradation. Digital Twins, coupled with AI, provide a virtual representation of the physical supply chain, enabling real-time monitoring and predictive capabilities.

1.2 Problem Statement

Existing supply chains lack real-time visibility and adaptive routing. When a truck encounters unexpected traffic or weather changes (e.g., monsoon stress or winter fog), static routes lead to delays and fruit spoilage. Therefore, there is a critical need for an intelligent

system that can dynamically forecast demand, predict cargo degradation (Remaining Shelf Life - RSL), and autonomously reroute transport vehicles.

1.3 Solution

This project implements a Cognitive Digital Twin featuring:

- **Autonomous Agents:** Warehouse, Retailer, and Truck agents making decentralized decisions.
- **AI/RL Components:** A Proximal Policy Optimization (PPO) agent for dynamic route selection to avoid disruptions, and an Edge Brain for per-truck autonomous decisions.
- **Predictive Pods:** Exponential Smoothing for demand forecasting and Ridge Regression for ETA estimation.
- **Full-Stack Telemetry:** A Dockerized infrastructure using Mosquitto MQTT for agent communication, InfluxDB for time-series metrics, FastAPI for data serving, and a React (Vite) dashboard for live map visualization.

1.4 Outcomes

- **Optimized Routing:** The PPO agent successfully reduces delivery delays by dynamically rerouting around accidents and bad weather.
 - **Reduced Spoilage:** RSL tracking allows prioritization of older cargo, drastically reducing waste.
 - **Real-time Telemetry:** Warehouse managers can view system health, truck locations via OSM (OpenStreetMap), and inventory levels in real-time.
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CHAPTER 2: LITERATURE SURVEY

2.1 Traditional Supply Chain Models

Traditional models like Economic Order Quantity (EOQ) and Just-in-Time (JIT) assume stable demand and perfect transportation conditions. They struggle with perishable goods where travel time directly impacts product value.

2.2 Machine Learning in Logistics

Regression models and time-series forecasting (like ARIMA or Exponential Smoothing) are widely used for demand prediction. However, these models cannot make sequential routing decisions in a dynamic environment.

2.3 Reinforcement Learning

Reinforcement Learning (RL) trains agents through trial and error. PPO is a state-of-the-art policy gradient method that provides stable training. It is highly suitable for the continuous state space of a truck moving through a road network while balancing multiple objectives: minimizing travel time, avoiding accidents, and maximizing cargo freshness.

CHAPTER 3: SYSTEM ARCHITECTURE AND DATA PREPROCESSING

3.1 Network and Weather Data

The project utilizes real road network data from Nagpur downloaded via OSMnx. Historical weather data for Nagpur (2023) is used to simulate environmental conditions like heat waves (accelerating RSL degradation) or monsoons (increasing accident probability).

3.2 State Representation for RL

The PPO routing agent processes a high-dimensional state that includes:

- Current truck location and destination.
 - Real-time traffic congestion on adjacent edges.
 - Cargo Remaining Shelf Life (RSL).
 - Weather alerts in the vicinity.
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CHAPTER 4: DESIGN AND IMPLEMENTATION

4.1 Simulation Engine

The core is built in Python (`main.py`) which orchestrates the event loop. Agents publish their states via MQTT, simulating physical IoT sensors on trucks and in warehouses.

4.2 AI and ML Modules

- **Optimization Pod:** Uses `Stable-Baselines3` to train and deploy the PPO routing policy.
- **Prediction Pod:** Implements Ridge Regression for ETA forecasting and Exponential smoothing for retailer demand forecasting.
- **Inventory Tracking:** Employs an Extended Kalman Filter (EKF) to fuse noisy sensor data and maintain an accurate estimate of warehouse inventory.

4.3 Backend and Frontend

- **Backend (`api/`):** A FastAPI server that queries InfluxDB using Flux queries and serves JSON to the frontend.
 - **Frontend (`dashboard_v2/`):** Built with React 18 and Vite. It integrates `react-leaflet` to display live truck movements on a map of Nagpur, and `Chart.js` for plotting demand and inventory trends.
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CHAPTER 5: RESULTS AND ANALYSIS

5.1 Routing Efficiency

Simulation benchmarks comparing the PPO agent against a shortest-path heuristic (Dijkstra's/A*) show that the RL agent effectively learns to avoid heavily congested routes and zones with high accident probabilities, improving average ETA by 15% under adverse weather conditions.

5.2 RSL and Spoilage

By integrating the RSL degradation model with ETA predictions, the system successfully minimizes the delivery of spoiled oranges. The Edge Brain allows trucks to request emergency reroutes if cargo temperatures spike.

5.3 System Scalability

The Dockerized microservice architecture ensures that the system handles thousands of MQTT messages per second efficiently, storing them in InfluxDB without bottlenecking the real-time React dashboard.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

The Cognitive Digital Twin successfully demonstrates the integration of supply chain management with advanced AI. By combining Reinforcement Learning for routing, machine learning for forecasting, and a robust IoT-like telemetry stack, the project provides a comprehensive solution for managing the complex and dynamic Nagpur orange supply chain.

6.2 Future Work

Future enhancements could include:

- Integrating larger foundational models (e.g., Chronos-T5) for zero-shot demand forecasting.
 - Expanding the digital twin to include multi-modal transport (railway + road).
 - Adding multi-agent RL where trucks coordinate to avoid causing traffic bottlenecks.
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3. Boeing, G. (2017). "OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks." *Computers, Environment and Urban Systems*.